

# GRANT NEILSON

Bioinformatician with seven years of industry experience analyzing multiple NGS data types, with a specialism in single cell RNA & whole genome sequencing. Highly proficient in R and Python, with a strong track record of developing analytical tools and packages for high-dimensional omics data. Passionate about building robust analytical workflows and contributing to open-source repositories to drive forward scientific research. An articulate and reliable individual with excellent communication skills and a strong track record of effective collaboration across multidisciplinary teams.

View this CV [online](#)

## CONTACT

✉ [grant.neilson5@gmail.com](mailto:grant.neilson5@gmail.com)

🐙 [GitHub](#)

in [LinkedIn](#)

## INDUSTRY EXPERIENCE

2026  
|  
2023

### Senior Computational Biologist

[CoSyne Therapeutics](#)

📍 London, UK

- Designed and implemented the end-to-end bioinformatic analysis for high-throughput single cell Mosaic drug screening, deploying hyperscale cloud infrastructure to run alignment and quality-control pipelines for ~3 million cells. Designed and implemented scalable pipelines for cell demultiplexing, sensitivity profiling of cell lines, and multi-omic biomarker identification.
- Drove biomarker discovery in glioblastoma by conducting WGS analysis on 80 primary tumour-derived cell lines, delivering comprehensive genetic characterisation (SNVs, SVs, CNVs). Integrated genomic profiles with CRISPRi screen data using advanced statistical models to uncover novel genetic dependencies and biomarker-target pairs.
- Executed single-cell CNV (sc-CNV) calling on proprietary perturb scRNA-seq data, identifying sub-clonal populations and contributing to the target identification platform.
- Recognized as a named contributor in version 3.4.1 of the NF-core Sarek pipeline. Optimised the running of the pipeline, achieving a 50% cost reduction and 30% decrease in computational time per sample.
- Internal R&D: Responsible for developing an internal container system using Docker and ECR. Author and Lead developer on several internal R and Python packages with responsibilities for maintenance. Established best practices for code development and review. Designed the templating system for all internal projects & packages using cruft.

2023  
|  
2020

### Computational Biologist

[Fios Genomics](#)

📍 Edinburgh, UK

- Directed projects focusing on high-dimensional omics data types (WES, WGS, scRNA-seq, proteomics), delivering comprehensive analyses and reports for clients.
- Developed in-depth knowledge of statistical methodologies, mastering model selection and fitting techniques, including SVM, elastic net, random forest, and linear/logistic regression, while accounting for covariates.
- Actively contributed to internal R&D by developing and maintaining specialized R packages, enhancing the team's analytical capabilities.
- Led client calls, effectively communicating complex analytical results, ensuring clarity and client understanding.

## LANGUAGES

📊 R  
🐍 Python  
🔗 Bash  
🔧 Nextflow

## TECHNOLOGIES

🐙 GitLab/GitHub  
🐳 Docker  
aws aws  
🖥️ Azure

## OPEN SOURCE & PUBLIC PROJECTS

📊 [bixverse](#)  
📊 [mainfoldsR](#)  
📊 [cosyne-cnutils](#)  
🔧 [Sarek](#)

Made with the R package [pagedown](#).

The source code is available on [GitHub](#).

Last updated February 2026.

2020  
|  
2019

## Computational Biologist

Complex Diseases Epigenetics group, Exeter Medical School

📍 Exeter, UK

- Developed novel bioinformatics pipelines for Epigenome/Genome wide association studies, mQTL analysis and quality control of Illumina EPIC array data.
- Took the lead on a project performing EWAS and GWAS analysis in ADHD patients. This involved fitting linear and logistic regression models while accounting for covariates.
- Was selected for a secondment to New York to work at the Mount Sinai medical school as a visiting bioinformatician. This was unfortunately cancelled due to COVID-19.



## EDUCATION

2019  
|  
2015

### MSCi Biology

University of Southampton

📍 Southampton, UK

- Grade : First Class Honors
- Winner of the British Society for Immunology Graduate of the year award.
- Recipient of the summer studentship from the Genetics Society (2017).

2015  
|  
2013

### School

Merchiston Castle School

📍 Edinburgh, UK

- A Levels: Biology, Chemistry, Maths



## SELECTED PUBLICATIONS

2025

### TP53 and RB1 are predictive genetic biomarkers for sensitivity to cytarabine in gliomas

BioRxiv

- L. Laaniste, J. Luginbuehl, M. Trinh, R. Watkins, G. Lueg, G. Neilson, et al.
- doi: <https://doi.org/10.1101/2025.11.20.688938>

2025

### Multi modal single cell foundation models via dynamic token adaptation

BioRxiv

- W. Zhao, A. Solaguren-Beascoa, G. Neilson, et al.
- doi: <https://doi.org/10.1101/2025.04.17.649387>

2025

### Detecting cell level transcriptomic changes of Perturb seq using Contrastive Fine tuning of Single Cell Foundation Models

BioRxiv

- W. Zhao, A. Solaguren-Beascoa, G. Neilson, et al.
- doi: <https://doi.org/10.1101/2025.04.17.649395>